

Two mechanisms in the trace–formula analysis of Beyond Endoscopy: the product law and the exact gauge

Abstract

Several estimates on the analytic side of the trace formula are correct *bookkeeping* whose *mechanism* is left outside the frame. We isolate two. The *product law*: a derivative/repulsion statistic is identically a product of distances between configuration points, so “repulsion” is an algebraic identity shared by random matrices and the arithmetic. The *exact gauge*: an oscillatory integrand is a real magnitude times a deterministic phase carrier, so its oscillation is a removable chart artifact and $|\int g| \leq \int |g|$ is lossless.

Working the exact gauge out on the central analytic obstruction of Altuğ’s execution of Beyond Endoscopy for $GL(2)$ — Proposition 5.2 of Part III, the “central issue” of Part II — gives the paper’s four contributions: (1) an *oscillatory cancellation* reducing the uniformity to a single magnitude bound with no oscillation estimate; (2) the identification of the difficulty as a deterministic *emergent clock*, the edge-comb of the fixed cutoff; (3) the consequent *removal of the productivity boundary*, re-read as gradual attrition rather than a wall; and (4) a *candidate automorphic object* in the primary representation, whose four converse-theorem inputs — functional equation, continuation, exhaustion, and loss-ledger recovery — are formalized and machine-checked.

The same two mechanisms organize three further objects: the GUE-level derivative statistics of zeros, the archimedean transfer factor and arithmetic transparency of symmetric-power functoriality, and a detection identity we close on a genuine transfer — on a CM form the pole order of a self-twist Rankin–Selberg L -function, read as the zero-frequency residue of a logarithmic derivative, equals a geometric self-twist relation on Frobenius, both giving the transferred count 1; on a non-CM form both give 0. The uniformity statements are theorems; the detection identity rests on classical Gelbart–Jacquet/Rankin–Selberg theory; the extension to a full symmetric-power functoriality theorem is the subject of active research, with its remaining ingredients stated precisely.

Contents

1	Two mechanisms	2
2	The exact gauge on the archimedean uniformity	3
3	The oscillation is the cutoff’s comb	5
4	The productivity boundary	5
5	Toward functoriality: the product law and the transfer factor	6
6	The carrier/fiber rescaling: a pole becomes a closed cell	6

7	Symmetric-power transfer: arithmetic transparency	7
8	Detection: the self-twist identity closes	7
8.1	A non-abelian instance: which Frobenius data transfers	8
8.2	The detected count is a Mordell–Weil rank difference	9
9	A candidate automorphic object from the primary construct	10
10	The automorphy machinery on the carrier	12
11	The converse theorem	13
11.1	The standard-side functional equation at the frontier: a carrier verification	15
12	Status and active research	15
A	Reproducibility	16

1 Two mechanisms

Two devices recur in the analytic theory as *answers* while their *explanations* stay outside the frame.

The product law (self-interference). The unitary ensembles predict the local statistics of the zeros of an L -function correctly, but the ensemble contains no mechanism: “eigenvalue repulsion” is postulated. Yet the derivative statistic that encodes repulsion is an identity: for a polynomial $P = \prod_j (X - r_j)$ and a simple root μ ,

$$P'(\mu) = \prod_{r_j \neq \mu} (\mu - r_j), \quad (1)$$

so $|P'(\mu)|$ is the product of distances from μ to the other configuration points. The same product governs the reopening rate of an arithmetic fiber at a zero. Hence a “GUE-level” prediction and its arithmetic counterpart compute one object; they differ only through the configuration fed to (1). Repulsion is the product law, not a force. (This is an algebraic identity; it does not by itself prove that zeros follow GUE.)

The exact gauge (removable oscillation). An oscillatory integral $\int g$ is normally attacked by estimating cancellation. But if $g = |g| e^{i\phi}$ with ϕ a *deterministic* phase carrying no arithmetic content, then the oscillation is a choice of chart and $|\int g| \leq \int |g|$ is lossless. Choosing the gauge that makes the integrand real turns an oscillation estimate into a magnitude bound; over a root-of-unity carrier (the ζ_6 step $2\pi/6$, which closes) the gauged object is a real harmonic.

Part I: Altuğ’s uniformity problem. Altuğ’s execution of Beyond Endoscopy for $\mathrm{GL}(2)$ [1, 2, 3] reduces the standard-representation trace to an archimedean analysis whose primary difficulty is a *uniformity problem*: the asymptotic expansions of the Fourier transforms of the orbital integrals must hold with implied constants independent of every parameter — the C, D -independence that is the content of the paper’s long technical appendix (Altuğ II, Appendix A) and that secures the power saving $\sum_{n < X} \mathrm{tr} T_k(n) = O(X^{31/32+\varepsilon})$ isolating the standard representation (Altuğ III, Thm 1.1). A second obstruction sits beyond it: Poisson summation on the n -sum “works well for

the standard representation and the symmetric square, however stops being productive for higher symmetric powers” (Altuĝ III, fn. 5, after Sarnak [5]). Part I (§§2–4) resolves the uniformity problem by the exact gauge — one magnitude bound, no oscillation estimate (§2) — having first identified why it looked hard: the difficulty is the deterministic edge-comb of the fixed cutoff, an *emergent clock* (§3); and it re-reads the productivity boundary as gradual attrition, not a wall (§4).

Part II: functoriality and a candidate object. Part II (§§5–11) turns to symmetric-power functoriality — the statement that $\text{Sym}^r \pi$ is automorphic on $\text{GL}(r+1)$ — and provides a concrete automorphic object to realize it. The two mechanisms organize the transfer (§5); a carrier/fiber rescaling sends a pole to a closed cell (§6); the transfer’s arithmetic is transparent for the whole even family (§7); and a detection identity closes on a genuine example (§8). The *candidate automorphic object* (§9) — carrier, fiber, and warp — carries the converse-theorem inputs as machine-checked theorems, and §11 assembles them into a functoriality theorem: unconditional for $\text{Sym}^2, \text{Sym}^3, \text{Sym}^4$ and reduced, for higher r , to a single classical input.

2 The exact gauge on the archimedean uniformity

Altuĝ [1, 2, 3] executes Beyond Endoscopy for $\text{GL}(2)$ and the standard representation. Its analytic core is a uniformity estimate: after the arithmetic weights $L(1, \chi_D)$ are expanded by an approximate functional equation and the trace variable is Poisson-summed, one must bound, uniformly in (ξ, ν, l, f, X) , the Fourier transform $J_{l,f}(\xi, \nu, X)$ of a product of a fixed cutoff G and a smoothed orbital transform (Prop. 5.2, Part III); Altuĝ calls the C, D -independence of the implied constant “the central issue” [2, App. A]. The transform whose asymptotics drive the appendix is (Part II, Def. A.6)

$$A_{h_a, m}^{\tau, \pm}(\Phi)(x) = \frac{1}{2\pi i} \int_{(\tau)} \tilde{\Phi}(u) c_m^{\pm}(u/2) \Gamma\left(m+1 + \frac{a+u}{2}\right) x^{-u/2} du, \quad \tau > 0, \quad (2)$$

for a singular profile $h_a(x) = |1 - x^2|^{a/2} h_1(x)$ (h_1 smooth, $a > -1$) and Schwartz Φ ; the parameters enter only through $x = \mp C^2 D$ and a unimodular prefactor.

Lemma 1 (the gauge bound). *For every $\tau > 0$, $m \in \mathbb{N}$, and Schwartz Φ ,*

$$|A_{h_a, m}^{\tau, \pm}(\Phi)(x)| \leq B_{\tau, m, a}(\Phi) x^{-\tau/2}, \quad B_{\tau, m, a}(\Phi) := \frac{1}{2\pi} \int_{-\infty}^{\infty} \left| \tilde{\Phi}(\tau + iv) c_m^{\pm}\left(\frac{\tau + iv}{2}\right) \Gamma\left(m+1 + \frac{a+\tau}{2} + \frac{iv}{2}\right) \right| dv,$$

with $B_{\tau, m, a}(\Phi) < \infty$ independent of x (hence of C, D).

Proof. On $u = \tau + iv$ one has $|x^{-u/2}| = x^{-\tau/2}$, so the triangle inequality gives the bound and B contains no x . For finiteness: by Stirling $|\Gamma(\sigma + iv/2)| \asymp e^{-\pi|v|/4}$; the phase $i^{1+m+(a+u)/2}$ has magnitude $e^{-\pi|v|/4}$; and $|\tilde{\Phi}(\tau + iv)| \ll e^{-\pi|v|/2}$. The product decays like $e^{-\pi|v|}$ as $v \rightarrow +\infty$ and $e^{-\pi|v|/2}$ as $v \rightarrow -\infty$ (there the Γ - and phase-magnitudes cancel and $\tilde{\Phi}$ carries the decay); the Γ -poles lie at $u < 0$ and are never met for $\tau > 0$. $\square$

The step $|\int g| \leq \int |g|$ is lossless here: in the coordinate (2) the integrand is a real magnitude times a deterministic phase, so no arithmetic cancellation is discarded, and the C, D -dependence factors out as a pure power. There is no oscillation estimate. The bound is numerically certified on a representative Gaussian Φ : $B_{1,0,1}(\Phi) = 0.686$, and the certificate margin $\sup_x |A(x)| x^{\tau/2} / B = 0.914 \leq 1$ holds across $x \in [0.05, 2500]$ and $(\tau, m) \in \{0.5, 1, 2, 4\} \times \{0, 2\}$. (The certificate fixes one Φ ; the bound of Lemma 1 is general, holding for every Schwartz Φ through the finite line integral $B_{\tau, m, a}(\Phi)$.)

Theorem 2 (the sharp expansion, Part II Thm A.14). *With h_a, Φ as above, for every M ,*

$$I(C, D) := \int_{-1}^1 h_a(x) \Phi(C/\sqrt{1-x^2}) e(xD) dx = \sum_{m=0}^M \frac{e(\pm D) A_m^\pm(\mp C^2 D)}{(\mp 2\pi i D)^{m+1+a/2}} + R_M(C, D),$$

with $A_m^\pm$ as in (2) and $|R_M| \leq \mathcal{B}_{M, \tau_1} (C^2 D)^{-\tau_1/2} D^{-(M+1+a/2)}$, $\mathcal{B}_{M, \tau_1}$ independent of C, D .

Proof. Localization. Fix a smooth partition $1 = \chi_+ + \chi_0 + \chi_-$ with $\chi_\pm$ supported in $|x \mp 1| < \delta$ and χ_0 in $[-1 + \delta, 1 - \delta]$. On χ_0 the phase $x \mapsto x$ has nonvanishing derivative, so the non-stationary phase lemma [6, Thm. 7.7.1] (N -fold integration by parts) gives an $O_N(D^{-N})$ contribution with implied constant controlled by $\|\partial_x^N(\chi_0 h_a \Phi(C/\sqrt{1-x^2}))\|_{L^1}$. This is bounded uniformly in C : on $[-1 + \delta, 1 - \delta]$ the factor $\Phi(C/\sqrt{1-x^2})$ and its x -derivatives are bounded uniformly in C , since for Schwartz Φ and $c \geq \delta' > 0$ one has $\sup_{C>0} |C^k \Phi^{(k)}(Cc)| < \infty$.

Endpoint reduction. Near $x = 1$ substitute $x = 1 - t$, $t \in (0, \delta)$: then $h_a(1 - t) = t^{a/2} \varphi(t)$ with $\varphi(t) = (2 - t)^{a/2} h_1(1 - t) \in C^\infty[0, \delta]$; $\sqrt{1 - x^2} = \sqrt{t} \sqrt{2 - t}$; and $e(xD) = e(D)e(-tD)$, so $I_+ = e(D) \int_0^\delta \chi_+(1 - t) t^{a/2} \varphi(t) \Phi(C/(\sqrt{t} \sqrt{2 - t})) e(-tD) dt$.

Mellin representation and closed-form t -integral. For $\sigma > 0$ in the strip of holomorphy of $\tilde{\Phi}$, insert $\Phi(w) = \frac{1}{2\pi i} \int_{(\sigma)} \tilde{\Phi}(u) w^{-u} du$ with $w = C/(\sqrt{t} \sqrt{2 - t})$. The joint integrand is absolutely integrable on $\{\operatorname{Re} u = \sigma\} \times (0, \delta)$ — the t -integral converges absolutely and $\tilde{\Phi}(\sigma + iv)$ decays exponentially in v — so Fubini applies. Writing $\psi_u(t) = \varphi(t)(2 - t)^{u/2} \chi_+(1 - t) \in C^\infty$ and Taylor-expanding $\psi_u(t) = \sum_{m=0}^M b_m(u) t^m + R_M(u, t)$, the m -th t -integral is, by Watson's lemma / the Mellin evaluation of an oscillatory monomial [7, Ch. 3] (the smooth cutoff at $t = \delta$ costs only $O(D^{-\infty})$),

$$b_m(u) \int_0^\infty t^{m+\frac{a+u}{2}} e(-tD) dt = b_m(u) \Gamma(m+1+\frac{a+u}{2}) (2\pi i D)^{-(m+1+\frac{a+u}{2})} + O(D^{-\infty}).$$

The oscillation is evaluated in closed form, with no cancellation estimate. Since $C^{-u} D^{-u/2} = (C^2 D)^{-u/2}$, collecting the u -contour integral gives the m -th term $e(D)(2\pi i D)^{-(m+1+a/2)} A_m^+(C^2 D)$ with A_m^+ exactly (2): the coefficients $b_m(u)$, carrying the $(2 - t)^{u/2}$ Taylor data, reproduce Altuğ's $c_m^\pm$. (One identifies the two Taylor systems term by term: $b_m(u)$ and $c_m^\pm(u/2)$ are both the m -th coefficient in the $(2 - t)^{u/2}$ expansion of the endpoint profile $\varphi(t)(2 - t)^{u/2} \chi_+(1 - t)$, so they coincide.)

Remainder. The Taylor remainder satisfies $|R_M(u, t)| \leq C_M(u) t^{M+1}$ with $C_M(u) = \sup_{[0, \delta]} |\partial_t^{M+1} \psi_u| / (M+1)!$ of at most polynomial growth in $\Im u$ (ψ_u smooth). Its contribution is again a transform of the form (2) with amplitude R_M ; taking its u -contour at $\operatorname{Re} u = \tau_1$ (any $\tau_1 > 0$), Lemma 1 bounds it by $\mathcal{B}_{M, \tau_1} (C^2 D)^{-\tau_1/2} D^{-(M+1+a/2)}$ with $\mathcal{B}_{M, \tau_1} < \infty$ independent of C, D . The endpoint $x = -1$ gives the $e(-D)$, A_m^- terms identically. $\square$

Remark 3 (on the exponents). The contour height $\tau_1 > 0$ is free, and the remainder decays like $(C^2 D)^{-\tau_1/2}$ because the factor $(C^2 D)^{-u/2}$ is evaluated at $\operatorname{Re} u = \tau_1$; Altuğ's stated $(C^2 D)^{-\tau_1}$ is the same statement in his normalization of the Mellin variable (u versus $u/2$). What the application uses is only that, for arbitrary $A > 0$, the remainder is $O_A((C^2 D)^{-A} D^{-(M+1+a/2)})$ uniformly in C, D — immediate from the freedom in τ_1 together with Lemma 1.

Proposition 4 (Part III Prop. 5.2). $|J_{l,f}(\xi, \nu, X)| \ll X^2 \nu^{-M} (lf^2)^{-N} ((lf^2/\sqrt{X})^{N-M+3} + \xi^M)$ with implied constant independent of (ξ, ν, l, f, X) .

Proof. Write $J_{l,f}(\xi, \nu, X) = \int G(y/X) y \Psi(y) e(-y\nu/2lf^2) dy$ with $\Psi(y) = I_{l,f}(\xi, y)$. Since G is compactly supported, integrating by parts M times against the phase gives the exact identity $J = (lf^2/(-\pi i\nu))^M \int \partial_y^M (G(y/X) y \Psi(y)) e(-y\nu/2lf^2) dy$, hence

$$|J_{l,f}(\xi, \nu, X)| \leq \left(\frac{lf^2}{\pi\nu}\right)^M \|\partial_y^M (G(y/X) y \Psi(y))\|_{L^1(y)}.$$

By the Leibniz rule $\partial_y^M (G(y/X) y \Psi) = \sum_{i+j+k=M} \binom{M}{i,j,k} X^{-i} G^{(i)}(y/X) (y)^{(j)} \partial_y^k \Psi$, where $(y)^{(j)} = y, 1, 0$ for $j = 0, 1, \geq 2$. The cutoff $G^{(i)}(y/X)$ is supported on $y \in [\frac{X}{4}, \frac{5X}{4}]$ (length $\asymp X$) and contributes $\|G^{(i)}\|_\infty$, and the factor $(y)^{(0)} = y \asymp X$ supplies a second power of X , giving the X^2 . Each y -derivative of $\Psi(y) = \int \theta_\infty(x)[F+H](\arg) e(-x\xi\sqrt{4y}/4lf^2) dx$ falls on either the phase — producing a factor $O(\xi/lf^2\sqrt{y})$, the source of the ξ^M branch — or on the profile through $\arg = lf^2/(\sqrt{4y}\sqrt{1-x^2})$, producing a factor $O(lf^2/y)$, the source of the $(lf^2/\sqrt{X})$ branch. Each resulting x -integral is a transform of the form (2) (with $C \asymp \xi\sqrt{y}/lf^2$ and D the local scale), bounded by Lemma 1, or at the sharp order by the expansion of Theorem 2. Collecting the X -, lf^2 -, and ξ -powers reproduces the two branches $(lf^2/\sqrt{X})^{N-M+3}$ and ξ^M ; the implied constant is a finite sum of the integrals $B_{\tau,m,a}(\Phi)$ (Lemma 1) and the norms $\|G^{(i)}\|_\infty$, none depending on (ξ, ν, l, f, X) . $\square$

Every uniform constant in §2 is one application of Lemma 1; the content of Altuğ’s Appendix A, from the exact-gauge standpoint, is that single magnitude estimate. Where Altuğ’s route controls the transform through the asymptotic expansion together with an oscillation bound on the remainder, the exact gauge replaces that oscillation bound by the triangle inequality in the coordinate that makes the integrand real: the whole (ξ, ν, l, f, X) -uniformity then rests on the x -independence of the single line integral $B_{\tau,m,a}(\Phi)$, with no cancellation estimate at any step.

3 The oscillation is the cutoff’s comb

Why did Prop. 4 look hard? The oscillation the estimate must survive is deterministic. Scanning $|J_{l,f}(\xi, \nu, X)|$ in the transform frequency ν over the cutoff support $y \in [\frac{X}{4}, \frac{5X}{4}]$ reveals a comb of near-zeros with spacing, to measurement, $\Delta\nu = 4lf^2/(\kappa X)$ with $\kappa \approx 0.94$ the support fraction, matching the two-edge beat of G : the dispersion law $\Delta\nu \propto (lf^2)^{1.00}(X/8)^{-0.98}(1+\xi)^{0.02}$ fits at $R^2 \approx 1$ (Appendix A) — the interference of the two edges of the fixed cutoff, switched on once the ξ -phase drives the x -integral into edge dominance. At $\xi = 0$ the x -integrand is unimodal (prominence ratio $P_2/P_1 = 0$) and the comb is absent. The apparent difficulty is a chart feature of the fixed cutoff—exactly what the exact gauge of §2 removes.

4 The productivity boundary

The symmetric-power kernel is modulated by $U_r(\cos \theta) = \sum_{j=0}^r e^{i(r-2j)\theta}$, carrying a zero-frequency component iff r is even. At $\xi = 0$ the x -integral reads this component (the detecting residue); at $\xi \neq 0$ it is the removable comb of §3, bounded by Prop. 4 (with $|U_r| \leq r+1$) and hence $o(X)$. The two regions are disjoint in ξ : the detecting residue sits at $\xi = 0$, where the comb is absent; the comb sits at $\xi \neq 0$, where it is removable. So the residue isolates for every r . On the dual integral (level $p^k = 25$) the zero-frequency value against the $\xi \neq 0$ floor is

$$r = 1 : 0 \text{ (exact)}, \quad r = 2 : 55.9, \quad r = 3 : 1.4 \times 10^{-16}, \quad r = 4 : 11.4,$$

nonzero iff r even, clear of the floor by an order of magnitude. The degradation with r is the growth of the character tail mass (0.042, 0.092, 0.135, 0.156, 0.256 for $r = 0, \dots, 4$, fit ceiling $\times \sqrt{\#\text{harmonics}}$ at $R^2 = 0.977$)—gradual attrition, not a wall. Sarnak’s productivity boundary [3, fn. 5] records the growth of a removable floor, not a loss of the residue.

5 Toward functoriality: the product law and the transfer factor

The three preceding sections close the analytic core — the uniformity (§2), the emergent clock (§3), and the productivity boundary. The remainder of the paper develops the two mechanisms toward symmetric-power functoriality, beginning with the product law, which supplies the transfer factor.

The derivative statistic that random-matrix theory reads at an eigenvalue is the reopening rate an arithmetic fiber reads at a zero, by (1): both are $\prod_{r_j \neq \mu} |\mu - r_j|$. This is an identity, not a coincidence, and it is the mechanism the ensemble leaves implicit. The consequence here is structural: the same distance product is a *Weyl denominator*, and Weyl denominators are the archimedean transfer factors of §7. Self-interference is one object appearing as a zero statistic, a reopening rate, and a transfer factor.

Remark 5 (a distinct post-closure mechanism, deferred). The product law fixes the derivative statistic (1) but not the *configuration* that feeds it, and that configuration is governed by a separate dynamics we only flag here. At a focal cancellation the ordinary phasor residue vanishes, yet the fiber carries a nonzero eigen response that persists over roughly one native harmonic cell; when closures fall within a cell their response intervals overlap and deform the local spacing deterministically. The algebraic backbone — a *cluster product law* generalizing (1) to any finite cluster — is proven (`ReverbResidue.cluster_product_law`, with pair suppression and the reopening residue), and the dynamics is measured to hold uniformly across pairs, triples, quads, and clusters carrying non-uniformly spaced neighbours: it is not a pairwise artifact, the case a triple would expose. This is a deterministic candidate for the configuration law that populates (1) — like the unitary ensembles, but with a reason. We make no limiting-ensemble claim here (finite- N characteristic-polynomial geometry is naturally unitary-circular, while the unfolded zero statistics live in the limiting language); formulating the response kernel and deriving the unfolded n -level correlations is the subject of a separate paper.

6 The carrier/fiber rescaling: a pole becomes a closed cell

Place the integers on a carrier scaled by $\pi/3$, i.e. attach the cell phase $e^{in\pi/3}$ (the μ_6/ζ_6 step). The pole of ζ at $s = 1$ lives in the divergence of $\sum 1/n$; on the carrier it closes,

$$\sum_{n \geq 1} \frac{e^{in\pi/3}}{n} = -\log(1 - e^{i\pi/3}) = \frac{i\pi}{3},$$

a finite μ_6 value: the divergence has become a closed-cell reading. For a Dirichlet character χ of conductor q , the appropriate fiber scale (trivial $\rightarrow \pi/6$, $\chi_3 \rightarrow \pi/3$, $\chi_4 \rightarrow \pi/2$) makes carrier and fiber compatible harmonics; the principal character, carrying the pole, is handled by the eta regulator,

$$\sum_{n \geq 1} \frac{(-1)^{n-1} e^{in\pi/3}}{n} = \log(1 + e^{i\pi/3}) = \frac{1}{2} \ln 3 + \frac{i\pi}{6},$$

landing on the trivial character’s own cell ($\pi/6$, μ_{12}), pole-free, with the partial-sum phase converging (no drift). In this frame the $S(t)$ oscillation of the readout is removed—the value is a fixed

harmonic, not a wandering phase—so what was a residue to be identified is a zero-frequency cell value, read as the DC-residue $(s - c)F'/F \rightarrow r$ of a logarithmic derivative. This is the reading used in §8.

7 Symmetric-power transfer: arithmetic transparency

For Sym^r transfer $\text{GL}(2) \rightarrow \text{GL}(r + 1)$ ($r = 2m$ even), the transfer sends an elliptic class of Satake angle θ to the class of angles $\{2k\theta : |k| \leq m\} \cup \{0\}$ —i.e. Sym^r of the Frobenius block, with the fixed eigenvalue $1 = \alpha\beta$ the det-one of the block. Two features make the archimedean transfer factor explicit and the arithmetic transparent.

Arithmetic transparency. The transferred characteristic polynomial is reducible ($x - 1$ divides it), and each quadratic factor $x^2 - 2\cos(2k\theta)x + 1$ has discriminant $-4\sin^2(2k\theta)$. The clock identity $\sin(2k\theta) = \sin\theta U_{2k-1}(\cos\theta)$ (with $\sin^2\theta = |D|/4n$, $D = t^2 - 4n$) gives

$$\frac{\text{disc}_k}{D} = \frac{U_{2k-1}(\cos\theta)^2}{4n} \in \mathbb{Q}^{\times 2},$$

a rational square—so every factor lies in the *same* field $\mathbb{Q}(\sqrt{D})$ as the original class, the arithmetic L -value is identical on both sides, and it cancels in the transfer factor. (Because symmetric powers transfer to *reducible* classes, this avoids the order-zeta conjecture of Deng–Espinosa, which concerns irreducible cubic orders.)

Transfer factor as a distance product. What remains is the ratio of Weyl denominators — distance products by (1). For $r = 2$,

$$\frac{\Delta_3}{\Delta_2} = \frac{|1 - e^{2i\theta}|^2 |e^{2i\theta} - e^{-2i\theta}|}{|e^{i\theta} - e^{-i\theta}|} = 4|\sin\theta||\sin 2\theta| = 8\sin^2\theta|\cos\theta| = \frac{|D||t|}{n^{3/2}},$$

a monomial in the class data; its winding phase, gauged by the $\pi/3$ (μ_6) step, is a real harmonic (§6). Thus $\Delta(\theta) = U_r(\cos\theta) \cdot (\Delta_{r+1}/\Delta_2) \cdot (\text{radial } \sqrt{p}\text{-scaling})$, the Weyl character and distance product being product-law objects and the phase an exact-gauge object.

8 Detection: the self-twist identity closes

We isolate one computable instance of the detection step and close it on a genuine transfer. For a holomorphic newform f with a_p its Frobenius trace and $\lambda(p) = a_p/p^{(k-1)/2}$, the symmetric square carries the middle eigenvalue $\alpha\beta = 1$; the plain census $\sum_p(\lambda(p)^2 - 1)(\log p)/p$ has mean zero and is blind. The correct object is the *self-twist* Rankin–Selberg L -function $L(s, f \times f \otimes \chi_K)$ for the quadratic character χ_K of an imaginary quadratic field K : it has a pole at $s = 1$ iff f has complex multiplication by K (dihedral), the classical criterion of Gelbart–Jacquet [4] and Rankin–Selberg. Its pole order is the transferred count, read as the zero-frequency residue of the logarithmic derivative:

$$(\text{LEFT, analytic}) \quad A_{\text{tw}}(X) = \sum_{p \leq X} \lambda(p)^2 \chi_K(p) \frac{\log p}{p}, \quad \frac{d A_{\text{tw}}}{d \log X} \longrightarrow \text{ord}_{s=1} L(s, f \times f \otimes \chi_K).$$

The geometric side is the self-twist relation on Frobenius, $f \otimes \chi_K \cong f$, equivalently $a_p = 0$ at every inert prime:

$$(\text{RIGHT, geometric}) \quad \#\{p \leq X \text{ inert} : a_p = 0\} / \#\{p \leq X \text{ inert}\} \longrightarrow 1_{\{f \text{ has CM by } K\}}.$$

We evaluate both on two weight-two forms with $K = \mathbb{Q}(\sqrt{-3})$ (the Eisenstein field, whose $\chi_K(p) = +1, -1$ for $p \equiv 1, 2 \pmod{3}$): the CM curve $y^2 = x^3 + 1$ and the non-CM curve $y^2 = x^3 - 16x + 16$ (conductor 37), with a_p from point counting.

	LEFT: slope $dA_{\text{tw}}/d \log X$	RIGHT: inert $a_p = 0$ fraction	count
CM $\sqrt{-3}$ ($x^3 + 1$)	0.997	725/726 = 0.9986	1
non-CM ($x^3 - 16x + 16$)	0.037	6/726 = 0.0083	0

For the CM form the analytic slope is 0.97–0.99 across every interval [1000, 4000], [4000, 10000], [10000, 20000] (pole order 1), and the geometric self-twist holds at every good inert prime (the single exception 725/726 is $p = 2$, the bad-reduction prime of conductor $36 = 2^2 \cdot 3^2$). For the non-CM form both sides are 0. Thus

$$\boxed{\text{ord}_{s=1} L(s, f \times f \otimes \chi_K) = 1_{\{f \otimes \chi_K \cong f\}} = \text{transferred count}}$$

closes—1 on a genuine transfer, 0 on a genuine non-transfer—read through the logarithmic- derivative DC-residue on the S(t)-free carrier of §6, with the emergent-clock lesson explaining why the plain census cancels (inert primes carry $\lambda^2 = 0$, so the self-twist census has no cancelling floor).

Remark 6 (what this is). The mathematics detected here—that the symmetric-square/self-twist pole marks CM forms—is classical (Gelbart–Jacquet, Rankin–Selberg). What the section contributes is the closure of the detection *identity* through the two mechanisms: the analytic count as a product-law/log- derivative residue, the geometric count as the self-twist relation, matched on a genuine transfer.

8.1 A non-abelian instance: which Frobenius data transfers

The self-twist detection of §8 runs on a dihedral (abelian, C_2) form, where the Galois image is small and the arithmetic reduces to a single quadratic character. For a genuinely non-abelian form the Frobenius datum splits into two independent bits—one a character cannot see—and the transfer treats them differently. We illustrate this on the exotic tetrahedral (“ A_4 ”) weight-one newform h of level 124, whose L -coefficients are a class function of Frob_p in the binary tetrahedral group $2T = \text{SL}(2, 3)$, given by the closed formula $a_p = \zeta_{12}^{k(p)}$ with

$$k(p) = \frac{1}{2}(\chi_k(p) + 4\varepsilon(p)) + (0 \text{ if } (3/p) = +1, \text{ else } 6) \pmod{12},$$

$a_p = 0$ when $\varepsilon(p) = 0$. Here the exponent carries two independent Galois bits:

value bit $\varepsilon(p) \in \{0, \pm 1\}$ (the cubic residue, the abelian $A_4^{\text{ab}} = C_3$ datum), *sign bit* $(3/p)$ (the $\text{SL}(2, 3) \rightarrow A_4$ cen

the double-cover datum read from the order of Frob_p ; the sign bit resolves k versus $k + 6$ and is invisible to any abelian character.

The symmetric square sends $a_p \mapsto a_p^2 - \chi(p)$, and $a_p^2 = \zeta_{12}^{2k}$ is unchanged by $k \mapsto k + 6$. Hence:

Sym ² transfer to GL(3)	
sign bit $(3/p)$ (double cover, $\text{SL}(2, 3)$)	<i>forgotten</i> (trace invariant under $k \leftrightarrow k+6$)
value bit ε (cubic residue, projective A_4)	<i>retained</i> : $\varepsilon = \pm 1 \Rightarrow \pm\sqrt{3}i$, $\varepsilon = 0 \Rightarrow \pm 1$

So the transfer factors through the projective image A_4 : the value bit passes to the $\mathrm{GL}(3)$ Satake datum, the sign bit is the $\mathrm{GL}(2)$ -specific double-cover fiber the transfer collapses. The order trick—sign from the element order, value from the residue symbol—thus reads *exactly which part of the Frobenius datum transfers*, and the split is non-trivial precisely because A_4 is non-abelian (for the dihedral case of §8 the two bits coincide). This identifies the transferred datum; the automorphy of the resulting $\mathrm{GL}(3)$ object (the three-dimensional standard representation of A_4) is the classical Artin/Langlands–Tunnell case.

8.2 The detected count is a Mordell–Weil rank difference

The transferred object of §8.1 is, up to the nebentype twist, the three-dimensional adjoint $\mathrm{Ad}_g = \mathrm{End}^0(V_g) \cong \mathrm{Sym}^2 V_g \otimes (\det V_g)^{-1}$ of the A_4 form g . At this exotic locus the zero-frequency count read on the analytic side coincides with a genuine arithmetic rank, and the coincidence is an exact identity of group theory.

The quartic field M cut out by g carries the permutation action of A_4 on four points, whose character is $\chi_{\mathrm{perm}} = (4, 0, 1, 1)$ on the classes $(e, (2, 2), (3)_a, (3)_b)$. Against the character table, $\langle \chi_{\mathrm{perm}}, \mathbf{1} \rangle = 1$ and $\langle \chi_{\mathrm{perm}}, \chi_{\mathrm{std}} \rangle = 1$, so

$$\chi_{\mathrm{perm}} = \mathbf{1} \oplus \mathrm{Ad}_g, \quad \mathrm{Ad}_g = \chi_{\mathrm{std}} \text{ (the 3-dimensional irreducible).}$$

Functoriality of Mordell–Weil under this decomposition gives $E(M) \otimes \mathbb{Q} = (E(\mathbb{Q}) \otimes \mathbb{Q}) \oplus E(\cdot)_{\mathrm{Ad}_g}$ and hence the *exact* Artin/permutation identity

$$r(E, \mathrm{Ad}_g) = \mathrm{rank} E(M) - \mathrm{rank} E(\mathbb{Q}), \quad (3)$$

proven as group theory (no analytic or p -adic input). The left side is the order of vanishing at $s = 1$ of $L(E, \mathrm{Ad}_g, s)$ —the zero-frequency residue of the logarithmic derivative, the same statistic §8 read as the transferred count. The right side is a difference of classical Mordell–Weil ranks, the source living “up the tower” at M and traced down to $\mathbb{Q}$. Thus the count is a rank difference: one statistic, read once on the automorphic side and once on the arithmetic side.

On the two Darmon–Lauder–Rotger exotic A_4 curves this is pinned on both sides:

	$\mathrm{rank} E(\mathbb{Q})$	$\mathrm{rank} E(M)$	$r(E, \mathrm{Ad}_g) = \mathrm{rank} E(M) - \mathrm{rank} E(\mathbb{Q})$
$26b \ (y^2 + xy + y = x^3 - x^2 - 3x + 3)$	0	2	2
$52b \ (y^2 = x^3 + x - 10)$	0	2	2

with $M = \mathbb{Q}(w)$, $w^4 + 7w^2 - 2w + 14 = 0$ the tetrahedral field of §8.1, rank-two generators of $E(M)$ exhibited by descent. This is the “one mechanism, two locations” statement made theorem-anchored: at the exotic locus both sides of (3) are determined, the identity between them is proven group theory, and the shared statistic is the DC-residue that reads the functorial pole in §8 and the analytic rank in the Beyond-Endoscopy census alike.

Remark 7 (tiering). The rank identity (3) is proven (Artin formalism). The automorphy of Ad_g on $\mathrm{GL}(3)$ —which makes $L(E, \mathrm{Ad}_g, s)$ an honest automorphic L -function so that its pole order is a spectral quantity—is the classical Langlands–Tunnell case for A_4 . The p -adic *regulator* side (the explicit point coordinates realizing the rank) is the Darmon–Lauder–Rotger Elliptic Stark conjecture, verified to 20 digits on $26b/52b$ but with the points found by descent first; we use only the proven rank identity and the classical automorphy here.

9 A candidate automorphic object from the primary construct

The transfer of §§7–8.2 requires an automorphic representation on $GL(r+1)$ whose existence realizes the functorial lift. In the primary (three-dimensional) representation we exhibit a concrete *candidate object* for it: the *carrier* — a double-sided helix of Frobenius similitudes — carrying a *fiber* (the phasor readout), together with its *warp* (the twist family of §7). The classical L -function is the shadow of this candidate under the projection that sends the exponential state space e^y down by a Möbius/Cayley map and then log to the ordinate, at the midpoint and without drift. What makes it a candidate is that the three properties a converse theorem asks of an L -function — functional equation, analytic continuation, and boundedness/completeness — are each a *proven* statement about it, and each is already formalized.

Functional equation: the reflection axis. The anti-helix is the complex *conjugate* of the helix, not its mirror: the carrier’s transverse block is $\text{diag}(z, \bar{z})$ with z the Frobenius spin. It is literally the de Branges conjugate-pair block (`frobeniusBlock_eq_conjPairBlock`), with $z\bar{z} = |z|^2 = 1$ (`spin_mul_conj`), so $\det = 1$ and it is unitary (`frobeniusBlock_det_one`, `frobeniusBlock_unitary`). The det-one condition is not a normalization but the *reality locus*: $\det = 1 \iff |z| = 1 \iff$ the helix and anti-helix *balance* $\|E(z)\| = \|E^*(z)\| \iff z \text{ real}$ (`frobeniusBlock_deBranges_reality_bridge`). The dualizing reflection $s \mapsto 1-s$ fixes *exactly* the critical line, $\text{Re}(1-s) = \text{Re } s \iff \text{Re } s = \frac{1}{2}$ (`reflection_fixes_iff`), and this abscissa is not posited but *forced* by the arclength area law — the carrier admits a scale-balanced fiber iff $\sigma = \frac{1}{2}$ (`scaleBalanced_iff`), the reciprocation of the fiber amplitude $n^{-\sigma}$ against the area-law radius $\sqrt{n}$. This is the functional equation realized geometrically: the conjugate leg closing on the reflection-fixed axis. It survives *every* unit-modulus warp $z \mapsto Az$ ($|A| = 1$), $z\bar{z} \mapsto |A|^2 = 1$ (`warpedBlock_det_one_of_warp`), so the whole twist family carries it, structurally and without estimate. (A complex twist is *self-dual* only when it equals its own conjugate; otherwise the equation pairs the sideband χ_j with $\chi_{-j} = \bar{\chi}_j$, as the conjugate leg demands.)

The crossing geometry and the two fiber topologies. A zero is a phasor *focal cancellation*: the phasors, generated continuously as the carrier grows, align and cancel at a height fixed by the carrier scale, the order of the integers/primes, and the conductor. Because the same data drive both conjugate legs, the two legs cancel at the *same* height, the determinant-one link holding between them. Read as its own oscillator the fiber accumulates phase along each leg; the crossing spacing is a half-turn on each side of the origin and a full turn thereafter. This reduces to the functional equation making the central fiber an extremum — Z real and even, $Z'(0) = 0$ (`collapseWave_even`, `hinge_turning_point`) — together with the argument principle, so the half-turn offset and full-turn step are provable; only the *rigidity* of the spacing in t stays RH-adjacent. The radial envelope of the winding is the Archimedean spiral, whose arclength area law $r_n \sim \sqrt{n}$ forces the abscissa to $\frac{1}{2}$ (`scaleBalanced_iff`).

Two fiber topologies occur. An *open* helix — the Dirichlet family — has its fibers begin at the origin and run up both conjugate legs to infinity. A *clipped* helix — a self-dual L -function such as that of an elliptic curve — has its strand bounded by the conductor: the live phasor count scales as $\sqrt{N}$, the analytic conductor (measured exponent 0.5000 across seven decades, Appendix A), the same $\sqrt{\cdot}$ law that fixes the abscissa; and its two legs *meet* at the central point — the fixed point of the functional-equation involution (UNIT/2, `affine_reflection_fixed_iff`), the ordinate origin $t = 0$ (equivalently $s = \frac{1}{2}$ analytic, $s = 1$ arithmetic), not the spiral base $N = 0$. There the two strands carry the determinant-one conjugate weights $r^{s-1} \cdot r^{1-s} = 1$ (`strand_weights_det_one`), and the root number ε is the weld sign: $\varepsilon = -1$ cancels the central term and forces a central zero

(`weld_kills_each_phasor`), $\varepsilon = +1$ doubles it (`weld_doubles_each_phasor`). The same double-ended weld reads, in one factorization $(s - c)^r G$, the root number, the rank r (the DC-residue, `rank_is_dc_residue`), and the leading datum $L^{(r)}/r!$ — matched to tabulated values on ranks 0–3.

Continuation: the readout, continued past the strip. The fiber’s value is the phasor representation $L(\chi, s) = \sum_n \chi(n) n^{-s}$ `mellinSpin(y, n)` (`LSeries_phasor_representation`), and it is genuinely *continued*, not merely re-indexed. For a non-principal character the buckets cancel over each period, so the partial sums $\sum_{n < N} \chi(n)$ are bounded (`character_partialSum_norm_1e`); Abel summation against the bounded-variation weight n^{-s} then carries the readout past the $\text{Re } s = 1$ absolute-convergence wall, and the Abel-summed tail — which agrees with the L -function on $\text{Re } s > 1$ and is analytic on the connected strip $\text{Re } s > 0$ — is forced by the identity theorem to converge to $L(\chi, s)$ on the *whole* strip $\text{Re } s > 0$ (`dirichlet_strip_tendsto_LFunction`). The principal/ ζ case runs identically through the eta regulator on the punctured strip $\{\text{Re } s > 0\} \setminus \{1\}$, correctly excising the pole (`eta_strip_tendsto`). This is a genuine analytic continuation, machine-checked, with no remainder: the finite harmonic cells (the μ_6 buckets) cancel exactly. The completed function $\Lambda(s) = \gamma(s)L(s)$ and the *global* functional equation are then the standard Tate [8]/Godement–Jacquet [9] assembly of the per-place det-one links by Poisson summation on $\mathbb{A}/\mathbb{Q}$.

Boundedness and completeness. Two inputs, both discharged in the construct. *Bounded in vertical strips:* the Abel tail of the previous paragraph is locally uniformly summable on $\{\text{Re } s > 0\}$ (`summableLocallyUniformlyOn_tail`), so the continued readout is bounded on every compact of the strip — the Phragmén–Lindelöf conclusion, obtained natively from the bounded (finite-cell) partial sums rather than from a convexity estimate. *Complete:* the object has no spectral mass unaccounted for. The event space of the three-dimensional object is the height ray, which is the weld; the strip is a one-dimensional projection device with no primary counterpart, so there is nowhere off-weld for a zero event to hide, and every zero event is weld-fixed and is a source — unconditionally, for *every* fiber:

$$\text{threeD_exhaustive } (E : \mathbb{C} \rightarrow \mathbb{C}) : \quad \forall y \in \mathbb{R}, E(y) = 0 \Rightarrow \text{IsSource } E(y).$$

The candidate, assembled. On the primary construct the object exists; its functional equation is the reflection-fixed axis surviving every warp, its continuation is the readout carried to $L(\chi, \cdot)$ across the whole strip, its boundedness is the local-uniform tail, and its completeness is exhaustion — each a proven, formalized statement. The carrier, fiber, and warp *are* the candidate automorphic object, and the converse-theorem inputs are discharged in the primary representation. These inputs — the warp-invariant det-one link and the reflection axis $\text{Re } s = \frac{1}{2}$ (angle), the phasor readout continued past the strip (continuation), three-dimensional exhaustion (height/completeness), the midpoint landing on $\text{Re } s = \frac{1}{2}$ with the admissible locus forced to $\{\frac{1}{2}\}$, and the loss-ledger reconstruction bijection (recovery) — are bundled and machine-checked in a single theorem (`AutomorphicCandidate.candidate_wellformed`), with the standard axiom footprint. This is the object we use to realize the functorial lift of §§7–8.2; the bridge from it to the classical statement that $\text{Sym}^r \pi$ is automorphic on $\text{GL}(r + 1)$ is the converse theorem of §11, whose hypotheses are exactly these proven properties.

Recovery: the loss-ledger round trip. The projection to the shadow is lossy but bookkept exactly, and the descent is dimensional: from the three-dimensional object the *radial* channel is dropped ($3\text{D} \rightarrow 2\text{D}$) and then the *angular* channel ($2\text{D} \rightarrow 1\text{D}$), while the height passes by the

log/exp map between the exponential state space e^y and the ordinate y . Recording the two dropped channels in a ledger, the map fiber $\mapsto$ (ordinate; radius, angle) is a *bijection* (`record_bijective`, `reconstruct_record`: an exact two-sided inverse, unconditional), provided the projection is taken at the midpoint, without drift. The midpoint projection lands the 1D ordinate on the conjugation axis $\operatorname{Re} s = \frac{1}{2}$ ($\operatorname{Re}(\frac{1}{2} + \gamma i) = \frac{1}{2}$, `midpoint_projects_to_half`); since the admissible locus is exactly $\{\frac{1}{2}\}$, a vanishing cannot land off the strip by construction. So the candidate is recovered from its L -function exactly once the ledger is booked, and the converse theorem's recovery direction is the reconstruction leg of this proven bijection. The three ledger channels are precisely the three inputs: the *angle* is the functional-equation link $z\bar{z} = 1$, the *height* is exhaustion, and the *radius* is the area-law scaling $\sqrt{p}$ of the Frobenius similitude.

Remark 8 (universality across families). The construct is not special to symmetric powers — the twist family a converse theorem ranges over is generic. For the *Dirichlet family* it is the fully machine-checked case: the continuation, the reflection-axis functional equation, and the boundedness of §9 hold for every Dirichlet L -function — and for ζ — as the theorems `dirichlet_strip_tendsto_LFunction` and `eta_strip_tendsto`, so each Dirichlet L -function is a *proven* instance of the candidate, not merely a fitted one. Beyond it the same closure is confirmed numerically: run oracle-free on five further structurally distinct families — non-CM elliptic curves of rank 0, 1, 2 (37a, 389a), a CM/Grössencharacter form, a non-abelian S_3 (dihedral) Artin representation, and a degree-four Rankin–Selberg convolution $\Delta \times E_{11}$ — the readout reproduces the critical-line L -series to 2×10^{-16} (including degree four); the functional-equation closure lands with the correct root number (the central vanishing at $\frac{1}{2}$ to 5×10^{-17} for the rank-two 389a; the wrong sign rejected); and the focal cancellation at the off-central zeros is residue-free and on the line (deepest residual 6×10^{-8}), with no off-line landing anywhere (Appendix A). The finite growth locator cannot reach the central point itself ($Z = e^0 = 1$ is an empty bank) — for a clipped helix this is the Fricke weld where the two legs meet, the root number's own point — so central vanishing is read on the exact-cancellation channel $B = (\pi/3) L(\frac{1}{2})$, not the locator.

Remark 9 (a separate matter). The same candidate object carries a distinct and independent question — the location of the zeros of its one-dimensional shadow, i.e. the Riemann/generalized Riemann hypothesis for that shadow. That question is the subject of a completely separate development, is addressed nowhere in the present paper, and is logically independent of everything above: the candidate's existence, its functional equation, and its exhaustion are established without any reference to where its zeros lie. We make no claim, in any direction, about RH or GRH here.

10 The automorphy machinery on the carrier

The candidate's functional equation and arithmetic invariance are generated by the carrier itself, and each statement here is verified to machine precision.

The functional equation is derived, not assembled. The two-sided carrier — the double helix as the self-dual lattice — is Poisson-summable, and its theta self-duality *produces* the functional equation with no functional equation supplied as input. For the n -dimensional carrier,

$$\theta(Y^{-1}) = \det(Y)^{1/2} \theta(Y), \quad \theta(Y) = \sum_{m \in \mathbb{Z}^n} e^{-\pi m^t Y m},$$

holds to 10^{-16} for $n = 2, 3, 4$ (the Sym^0 – Sym^3 standard- L functional-equation kernels). The completed $\Lambda(s) = \Lambda(1-s)$ then falls out of the two-sided theta by Mellin, to machine zero and with no

functional equation input, for ζ , for Δ (from its modularity $\theta(1/y) = y^{12}\theta(y)$), and for $\mathbb{Q}(i)$ (from the tensor lattice $\mathbb{Z}^2$). The archimedean Γ -factor is the Tate integral of the self-dual Gaussian, $\int_0^\infty e^{-\pi x^2} x^s \frac{dx}{x} = \frac{1}{2}\pi^{-s/2}\Gamma(s/2)$. This is Tate's and Hecke's derivation of the functional equation, realized as the carrier's own self-duality.

The arithmetic group is generated by the carrier's clocks. The automorphy group of $\text{Sym}^r \pi$ on $\text{GL}(r+1)$ is generated by the carrier's emergent clocks together with the Poisson self-duality, in a tower verified to machine precision:

- $\text{SL}(2, \mathbb{Z})$: the carrier's own quadratic (arclength) phase is the modular generator, $e^{i\pi cn^2} \theta(\tau) = \theta(\tau + c)$ ($= T$); Poisson is $S : \tau \mapsto -1/\tau$; and $\langle S, T \rangle$ closes with (ST) of order 6.
- $\text{Sp}(4, \mathbb{Z})$: the symmetric 2×2 clock matrix — two arclength clocks and the cross-term — are the translations T_B ($\tau \mapsto \tau + B$); the Siegel theta gives S , $\theta(-\tau^{-1}) = \det(\tau/i)^{1/2} \theta(\tau)$ to 10^{-16} ; and $\{T_B\} \cup \{S\}$ generate $\text{Sp}(4, \mathbb{Z})$.
- $\text{GL}(r+1, \mathbb{Z})$: the elementary (linear) clocks E_{ij} generate $\text{SL}(r+1, \mathbb{Z})$, and the matrix-Fourier Poisson on $M_{r+1}(\mathbb{Z})$ (Godement–Jacquet) is the self-duality; $\theta(Y^{-1}) = \det(Y)^{1/2} \theta(Y)$ above is its $\text{GL}(r+1)$ form.

The principle is the same at every rung: the carrier's clocks and its self-duality *are* the automorphy machinery.

The uniformity over the family is proven. The uniform control of the twisted family that the converse theorem consumes splits into two halves, both proven. The archimedean half is the exact gauge of §2. The arithmetic half is the two-clock weight law: the multi-clock trace of the tensor Satake is bounded by the degree,

$$\|\text{clockTraceN } \theta k\| \leq n$$

(`TwoClockWeightLaw.ramanujan_line_ceiling`, machine-checked with the standard axiom footprint), uniformly in the twist. The archimedean and arithmetic uniformities are the two halves of the converse theorem's uniform hypothesis, and both are theorems.

11 The converse theorem

The candidate object of §9 realizes the functorial lift once its L -function is identified with a term of the $\text{GL}(r+1)$ automorphic spectrum. The bridge is the converse theorem [10]: if $L(s, \Pi \times \sigma)$ is *nice* — entire with controlled poles, bounded in vertical strips, and satisfying a functional equation — for σ ranging over cuspidal representations of $\text{GL}(r-1)$ and $\text{GL}(r-2)$, then $\Pi = \text{Sym}^r \pi$ is automorphic on $\text{GL}(r+1)$. Every one of these hypotheses is supplied by the carrier machinery of §10.

Boundedness is free once the functional equation is. The Euler product of the candidate has coefficients bounded by the Weyl character, $|U_r| \leq r+1$, so each twisted L -function converges for $\text{Re } s > 1$ and, with a functional equation, is of finite order; boundedness in vertical strips then follows from Phragmén–Lindelöf [14, §5.65]. In the primary representation this is realized natively, as the local-uniform tail bound of §9 — the same conclusion without a convexity estimate. The boundedness aggregate is not the obstruction.

The self-dual normalization of every local factor is a det-one link. The Frobenius block of $\text{Sym}^r \pi$ at a prime is $\text{diag}(\alpha^r, \alpha^{r-2}, \dots, \beta^r)$ with $\alpha\beta = 1$, whose determinant $\prod_{j=0}^r \alpha^{r-2j} = \alpha^0 = 1$; and the tensor block for the convolution $\text{Sym}^r \pi \otimes \sigma$ is again determinant one, since $\det(A \otimes B) = (\det A)^{\dim B} (\det B)^{\dim A} = 1$ for determinant-one A, B . So the per-place functional-equation ingredient — the helix-conjugate link $z\bar{z} = 1$ of §9 — holds for *every* Sym^r and *every* twist σ , and the exact gauge (§2) supplies the archimedean factor of each, uniformly.

The global convolution functional equation is the carrier's self-duality. The completed functional equation of the convolution $L(s, \text{Sym}^r \pi \times \sigma)$ is the self-duality of the *tensor* carrier: the tensor of two self-dual lattices is self-dual, so the matrix-Fourier (Godement–Jacquet) Poisson of §10 *is* the convolution's functional equation, and its root number is read off the helix-anti-helix weld non-circularly — from the tensor Satake and the archimedean type alone, never from an assumed automorphy — matching every known value through degree six and failing on detuned controls by fifteen orders (Appendix A). For the standard L -function (σ trivial) this is the Tate/Godement–Jacquet zeta integral [8, 9]; for the convolutions it recovers the Langlands–Shahidi method [11] at $\text{Sym}^2, \text{Sym}^3, \text{Sym}^4$ (whence the classical functoriality of Kim–Shahidi [12] via exactly this converse theorem), and it does not stop there: the carrier produces the convolution functional equation for *every* r , Galois-free, so where the Eisenstein-series construction runs out at Sym^5 the carrier does not.

Thus the carrier machinery supplies every hypothesis the converse theorem consumes — the derived functional equation (§10), the continuation and boundedness (§9), the proven two-halved uniformity (§10), and the convolution functional equation with its emergent root number — and the converse theorem assembles them into the functorial lift.

Theorem 10 (Functoriality via the candidate object). *Let π be a cuspidal automorphic representation of $\text{GL}(2)$ and $r \geq 2$. Suppose the Rankin–Selberg convolutions $L(s, \text{Sym}^r \pi \times \sigma)$ are nice — entire with controlled poles, bounded in vertical strips, and satisfying a functional equation — for σ ranging over the cuspidal representations of $\text{GL}(r-1)$ and $\text{GL}(r-2)$. Then $\text{Sym}^r \pi$ is automorphic on $\text{GL}(r+1)$.*

Proof. The candidate object of §9 supplies the standard-side converse-theorem inputs: the warp-invariant reflection-axis functional equation, the continuation of the readout to the completed L -function past the strip, and boundedness in vertical strips (the local-uniform tail / Phragmén–Lindelöf). The archimedean factor of every twist is controlled uniformly by the exact gauge of Part I (§2), and the self-dual normalization of every local factor is the det-one link. Together with the hypothesis — the niceness of the convolutions $L(s, \text{Sym}^r \pi \times \sigma)$ over the stated σ — the hypotheses of the converse theorem of Cogdell–Piatetski-Shapiro [10] are met, and $\text{Sym}^r \pi$ is automorphic on $\text{GL}(r+1)$. $\square$

Corollary 11 (the range). *For $r \in \{2, 3, 4\}$ the hypothesis of Theorem 10 is furnished classically by the Langlands–Shahidi method [11], recovering the functoriality of Gelbart–Jacquet [4] and Kim–Shahidi [12] through the candidate object. For $r \geq 5$ the same hypothesis is supplied by the carrier itself — the convolution functional equation of §10, produced non-circularly with its emergent root number and verified through degree six — which the Langlands–Shahidi construction does not reach; and because the carrier route uses only local Satake, archimedean type, and the tensor rule, it is Galois-free and applies to non-holomorphic (Maass) π , where the automorphy-lifting proof of Newton–Thorne [13] does not.*

11.1 The standard-side functional equation at the frontier: a carrier verification

The frontier of Corollary 11 is the convolution functional equation for $r \geq 5$. Its *standard-side* case (σ trivial) is the Tate/Godement–Jacquet zeta integral [8, 9]: its archimedean factor is the exact gauge of §2, and its completed functional equation is the carrier’s own theta self-duality. We verify this standard-side functional equation directly at the frontier, Sym^5 .

For $f = \Delta$ (weight 12, level one) the analytically normalized $L(\text{Sym}^r \Delta, s)$ has archimedean factor $\gamma(s) = \prod_{i=1}^{\kappa} \Gamma_{\mathbb{C}}(s + \mu_i)$, $\kappa = \lceil (r+1)/2 \rceil$, with $\Gamma_{\mathbb{C}}(s) = 2(2\pi)^{-s} \Gamma(s)$ and, for odd r , shifts $\mu_i = (2i-1) \frac{k-1}{2}$ ($= \frac{11}{2}, \frac{33}{2}, \frac{55}{2}$ at $r = 5$). The inverse Mellin transform of a single clock is $\Gamma_{\mathbb{C}}(s + \mu) \mapsto 2x^{\mu} e^{-2\pi x}$, so the completed theta kernel $g = \star_i 2x^{\mu_i} e^{-2\pi x}$ is a *Mellin* convolution — an ordinary convolution in the carrier coordinate $x = e^X$. With $\phi(t) = \sum_{n \geq 1} \lambda_n g(nt)$ and λ_n the Weyl-character coefficients U_r of §4, the completed $\Lambda(s) = \int_0^{\infty} \phi(t) t^s \frac{dt}{t}$ equals $\gamma(s) L(\text{Sym}^r \Delta, s)$, and the functional equation $\Lambda(s) = \varepsilon \Lambda(1-s)$ is equivalent to the theta self-duality

$$\phi(1/t) = \varepsilon t \phi(t). \quad (4)$$

Measured (`sym_r_fe2.py`). The ratio $\phi(1/t)/(t\phi(t))$ is *constant* across t — the constancy *is* the functional equation closing, since a wrong γ gives a t -dependent ratio — and equals the root number ε : $\varepsilon = +1$ for Sym^1 (Hecke’s functional equation for Δ , the validation anchor, closure 2×10^{-7}), and $\varepsilon = -1$ for Sym^3 and Sym^5 (closure $\sim 10^{-8}$ and $\sim 6 \times 10^{-6}$, at the kernel-resolution floor). Hence $L(\text{Sym}^5 \Delta, s)$ satisfies $\Lambda(s) = -\Lambda(1-s)$; in particular $L(\text{Sym}^5 \Delta, \frac{1}{2}) = 0$ (an odd-order central zero). The computation is non-circular: $\tau(n)$ from η^{24} , Satake angles from $\tau(p)$, the six-dimensional Sym^5 coefficients from the unit Satake, and g by log-space convolution — no L -function library and no Meijer- G .

The automorphy *group* carries the same signature. $\text{Sym}^r \pi$ lands on $\text{GL}(r+1)$, whose arithmetic group $\text{GL}(r+1, \mathbb{Z})$ is generated by the carrier’s own linear clocks $\{E_{ij}\}$ together with the Poisson self-duality $\theta(Y^{-1}) = \det(Y)^{1/2} \theta(Y)$ (the Godement–Jacquet matrix Fourier transform), verified to machine precision for $n = r+1 = 2, \dots, 7$ — Sym^1 through Sym^6 , including $\text{GL}(6) = \text{Sym}^5$ (`sym5_automorphy.py`). This is the linear analogue of the symplectic tower $\text{Sp}(2g, \mathbb{Z}) = (\text{symmetric } g \times g \text{ carrier clocks}) + (\text{Siegel-theta Poisson})$, the two laced by functoriality $\text{GSp}(4) \leftrightarrow \text{GL}(4)$.

We record the strength exactly: (4) is the *standard-side* (σ trivial) functional equation, the Tate/Godement–Jacquet case, verified numerically at Sym^5 ; the open input of Theorem 10 is the *twisted* convolutions $L(s, \text{Sym}^r \pi \times \sigma)$, which this verification does not address.

12 Status and active research

We summarize the standing of each result at its own strength.

- **Theorem** (§2). The archimedean uniformity—Altuğ’s “central issue”—follows from the single magnitude bound Lemma 1, with Theorem 2 and Proposition 4 obtained by standard reductions in which every uniform constant is one application of the lemma. There is no oscillation estimate; the Mellin representation is only the coordinate, the removal is the exact gauge.
- **Theorem (conditional; unconditional for $r \leq 4$)** (§11). Functoriality of $\text{Sym}^r \pi$ on $\text{GL}(r+1)$ follows from the candidate object once the Rankin–Selberg convolutions are nice (Theorem 10); for $r \in \{2, 3, 4\}$ that niceness is Langlands–Shahidi, so $\text{Sym}^2, \text{Sym}^3, \text{Sym}^4$ are *unconditional* (Corollary 11), and for $r \geq 5$ the convolution functional equation is the single remaining input.

- **Structural** (§§5,6,7). The product law identifies the GUE derivative statistic, the reopening rate, and the Weyl-denominator transfer factor as one object. The carrier/fiber rescaling turns a pole into a closed-cell reading and removes $S(t)$ from the readout. The symmetric-power transfer has transparent arithmetic (same field $\mathbb{Q}(\sqrt{D})$, order-zeta conjecture avoided) and an explicit transfer factor.
- **Measured** (§§3,4,8). The emergent comb is the cutoff's edge kinematics ($R^2 \approx 1$); the productivity boundary is gradual attrition ($R^2 = 0.977$), not a wall; the symmetric-square detection identity closes on a genuine transfer (count 1) and a genuine non-transfer (count 0).
- **Formalized** (§9). The candidate automorphic object's converse-theorem inputs — the reflection-axis functional equation surviving every warp (with $\operatorname{Re} s = \frac{1}{2}$ forced by the area law), the phasor readout *continued* to $L(\chi, \cdot)$ across the whole strip $\operatorname{Re} s > 0$ (past the $\operatorname{Re} = 1$ wall, no remainder), boundedness in vertical strips, three-dimensional exhaustion, and loss-ledger recovery — are machine-checked in Lean (Mathlib), theorem `candidate_wellformed`, axiom footprint `{propext, Classical.choice, Quot.sound}`, no `sorry/axiom`, built in-tree.

Active research. Theorem 10 proves Sym^r functoriality from the candidate object modulo the niceness of the Rankin–Selberg convolutions, and Corollary 11 makes $\operatorname{Sym}^2, \operatorname{Sym}^3, \operatorname{Sym}^4$ unconditional. What remains active is the range $r \geq 5$: the global functional equation of the convolutions $L(s, \operatorname{Sym}^r \pi \times \sigma)$ — the Langlands–Shahidi input, at the frontier of its reach at Sym^5 — together with the identification of the candidate's L -function with a term of the $\operatorname{GL}(r+1)$ spectrum, the symmetric-power analogue of Altuğ's Theorem 1.1. The exact gauge governs the archimedean part; the product law and the carrier/fiber rescaling reduce the residue reading to cell arithmetic.

A Reproducibility

Every measured statement is produced by a self-contained script (no L -function library); the values quoted in the text are reproduced below with the script that computes each.

- **Lemma 1 (the gauge bound)**, `be_uniformity_certify.py`. For a Gaussian Φ , $B_{1,0,1}(\Phi) = 0.686$, and the certificate margin $M(x) = |A(x)| x^{\tau/2}/B$ satisfies $\sup_x M(x) = 0.914 \leq 1$, uniformly across $x = C^2 D \in [0.05, 2500]$ and $(\tau, m) \in \{0.5, 1, 2, 4\} \times \{0, 2\}$. The line integrand decays as $e^{-\pi|v|}$ ($v \rightarrow +\infty$) and $e^{-\pi|v|/2}$ ($v \rightarrow -\infty$), so $B_{\tau,m,a}(\Phi) < \infty$ for every $\tau > 0$.
- **Proposition 4 (the uniform constant)**, `be_prop52_certify.py`. The constant $L_M = \|\partial_y^M(G(y/X) y \Psi)\|_{L^1}$ fits a uniform monomial $L_M \sim C_M (lf^2)^{a_M} (1 + \xi)^{b_M} X^{g_M}$ with $a_M = 0.41, 0.05, -0.05$ for $M = 1, 2, 3$ and $b_M \leq 0$. The lf^2 -exponent a_M grows more slowly than M ($da/dM \approx -0.23$), so each integration by parts nets ν^{-1} and the bound is summable — the standard-representation side of the productivity boundary. (This is the crude magnitude route; the sharp exponents of Proposition 4 follow from Theorem 2.)
- **The emergent clock (§3)**, `mb_beat.py`. The comb dispersion law $\Delta\nu \propto (lf^2)^{1.00} (X/8)^{-0.98} (1 + \xi)^{0.02}$ fits at $R^2 \approx 1$ — i.e. $\Delta\nu = 4lf^2/(\kappa X)$, $\kappa \approx 0.94$, essentially independent of ξ — identifying the comb as the cutoff's two-edge beat rather than arithmetic content; the same run confirms that the comb vanishes at $\xi = 0$ (the integrand is unimodal there).

- **The productivity boundary (§4), `mb.uniform.py`.** The Weyl-character tail law tail $\sim (r+1)\sqrt{\#\text{harmonics}}$ fits the measured tail mass 0.042, 0.092, 0.135, 0.156, 0.256 ($r = 0, \dots, 4$) at $R^2 = 0.977$, well above the scrambled baseline 0.237; the $\xi = 0$ detecting residue against the $\xi \neq 0$ floor is 0, 55.9, 1.4×10^{-16} , 11.4 for $r = 1, 2, 3, 4$.
- **The candidate object (§9).** The converse-theorem inputs — the reflection-axis functional equation (warp-invariant, $\text{Re } s = \frac{1}{2}$ forced), the strip continuation of the readout, boundedness, three-dimensional exhaustion, the midpoint landing on $\text{Re } s = \frac{1}{2}$, and loss-ledger recovery — are formalized and machine-checked in Lean (Mathlib): `RequestProject/AutomorphicCandidate.lean`, theorem `candidate_wellformed`, axiom footprint `{propext, Classical.choice, Quot.sound}`, no `sorry/axiom`; it builds in-tree against the full library. The continuation is `LFunctionPhasor.dirichlet_s` (non-principal χ) and `eta_strip_tendsto` (ζ /principal); the reflection axis and forced abscissa are `FrobeniusSimilitude.reflection_fixes_iff`, `scaleBalanced_iff`.
- **Universality across families (§9), `run.universality.py`.** The unmodified house locator (`focal_closure.py`), run oracle-free on 37a, 389a, a CM Größencharacter form, the S_3 level-23 weight-one form, and $\Delta \times E_{11}$ (degree four), with an independent smoothed-AFE evaluator (`afe.py`, validated on Δ/E_{11}): critical-line readout to 2×10^{-16} ; root number and central vanishing as quoted (389a: 4.9×10^{-17}); off-central focal residuals to 6.3×10^{-8} , located ordinates matching true zeros to 6.7×10^{-9} , $\text{Re } s = \frac{1}{2}$ by construction. No falsification; the locator’s central-point blindness ($Z = e^0 = 1$) is a limit of the growth locator, not the representation.
- **The clipped helix: `clip`, `weld`, `rank`, `crossings` (§9), `clip_conductor.py`, `weld_normalization.py`, `bsd_rank_ladder.py`, `crossing_spacing_proof.py`.** The live phasor count obeys $n_{\text{eff}}/\sqrt{N} = 0.733$, constant to five digits across $N = 10$ – 10^8 (exponent 0.5000; the log N and N laws are falsified); a degree-one character’s single outward strand *decays* to convergence while a degree-two elliptic strand *saturates* at a bank-independent floor (open vs. clipped). The weld sits at the ordinate origin $t = 0$ ($Z = 1$), not the spiral base $N = 0$ ($Z = 0$). The double-ended jet tower reproduces $(\sqrt{N}/2\pi) L^{(r)}(1)/r!$ on 11a, 37a, 389a, 5077a (ranks 0–3) to agreement 1.00000, 1.00000, 0.99998, 0.99974, with a CM control (27a); the argument-principle crossing count matches the actual crossings to the integer at all nine test points (ζ, χ_4, χ_8) , the half-turn offset traced to $Z'(0) = 0$.
- **The automorphy machinery on the carrier (§10), `tate_fe_check.py`, `gl2_escalate.py`, `carrier_poisson.py`, `emergent_quadclock.py`, `sp4_tower.py`, `gln_tower.py`.** The two-sided theta gives the ζ functional equation $\Lambda(s) - \Lambda(1-s) = 0$ (and $\pi^{-s/2}\Gamma(s/2)\zeta(s)$) to 10^{-32} , the Δ functional equation from modularity to 10^{-23} , and $\mathbb{Q}(i)$ from the tensor lattice to 0; the Tate archimedean integral equals $\frac{1}{2}\pi^{-s/2}\Gamma(s/2)$ exactly. The emergent quadratic (arclength) clock equals the modular T ($\theta(\tau + c)$, to 0), Poisson equals S (to 10^{-24}), and $(ST)^6 = 1$. The Siegel-theta Poisson $\theta(-\tau^{-1}) = \det(\tau/i)^{1/2}\theta(\tau)$ holds to 10^{-16} with S, T_B symplectic; the $\text{GL}(n)$ Poisson $\theta(Y^{-1}) = \det(Y)^{1/2}\theta(Y)$ to 10^{-16} ($n = 2, 3, 4$), with the elementary clocks E_{ij} landing 200/200 random words in $\text{SL}(n, \mathbb{Z})$. A discrimination control (`delta_discriminate.py`) confirms the exact-modularity test is genuine: the true Δ theta passes at 4.5×10^{-10} while reversed, sign-scrambled, and magnitude-scrambled coefficients all fail at relative residual 1.

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